CHAPTER 3

Audit the Idea Before You Build It

How to turn every promise in a plan into a test, and decide what to keep before you spend money

In this chapter

- Why claims are cheap to write and expensive to believe
- The five-step claims-to-evidence method
- The evidence ladder: how much to trust a source
- A worked example: auditing fourteen claims in my own robotic grocery-cart concept
- Common traps, a worksheet, a checklist, and a free exercise

I wrote the claims in this chapter. That is the first thing you should know.

In 2025 I designed Glidecart, a hands-free grocery cart that follows a shopper through the store, scans items as they go into the basket, and answers questions such as “Where is the almond milk?” I drew it, named it, and wrote a business plan, an executive summary, a white paper, and a legal and compliance outline for it. Those documents said the cart would use quantum computing to cut shopping time by 20%, protect payments with quantum key distribution, and run on quantum chips. They said the cart was compliant with food-safety, workplace-safety, and accessibility rules. They projected revenue, return on investment, and labor savings.

Every one of those statements read well. None of them had been tested.

A year later, while studying IT auditing in graduate school, I learned a habit that changed how I read my own work. An auditor does not ask whether a statement sounds right. An auditor asks, *What evidence*

supports this, and how strong is it? When I asked that question of my Glidecart documents, claim by claim, about half of them fell apart. Some were contradicted by evidence. One rested on a simple multiplication mistake. One described a chip that turned out to be a touchscreen driver. A few were good ideas that simply had not been tested yet.

This chapter teaches you to do the same thing to your own ideas, your vendors' promises, and the products your organization is about to buy. It is the cheapest risk control in this book. It costs a few days of careful reading and, sometimes, an afternoon of running a free experiment. It can save you from building the wrong thing.

Claims Are Cheap; Evidence Is Cheaper Early

Anyone can write “reduces costs by 30%.” It takes one second. Checking that sentence takes much longer, and it gets more expensive the later you do it. Before you build, checking a claim means reading a datasheet or running a small test. After you build, it means a redesign. After you ship, it can mean a recall, a lawsuit, or a regulator.

Chapter 1 showed what this looks like in real products. Ring doorbells and Alexa speakers were not undone by brilliant hackers. They ran into trouble because of design decisions about who could see customer videos and how long voice recordings were kept, and the Federal Trade Commission held the companies responsible (Federal Trade Commission [FTC], 2023a, 2023b). Each of those decisions was once a sentence in a plan that nobody challenged.

Innovation documents are especially full of untested claims, for understandable reasons. They are written to persuade investors, partners, and judges. They borrow statistics from market reports. They name the newest technology because it sounds impressive. In my case, quantum computing was the impressive technology. In yours it might be artificial intelligence, blockchain, or “military-grade encryption.” The method in this chapter works the same way for all of them.

The auditor's question

Auditors call a statement made by management an *assertion*. “All terminated employees lose access within one day” is an assertion. The auditor's job is to collect enough evidence to say

whether it is true. Chapter 6 shows a script that tests exactly that assertion against real account records.

In this chapter, every sentence in a plan that promises a result, names a technology, or states compliance is treated as an assertion, including your own.

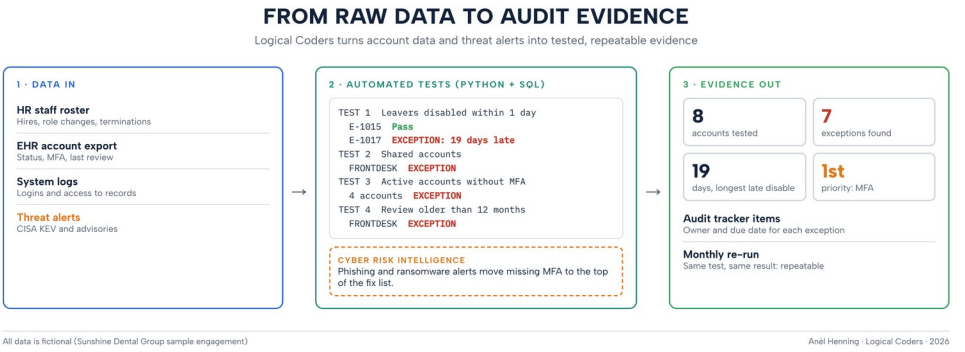


Figure 3.1 From raw data to audit evidence. In an audit, a claim becomes evidence only after data are collected, tested in a repeatable way, and turned into findings with owners and dates. The claims-to-evidence method applies the same discipline to a product idea. (Data shown are from the fictional Sunshine Dental Group case in Chapter 5.)

The Claims-to-Evidence Method

The method has five steps. You can do the first two in an afternoon with a highlighter.

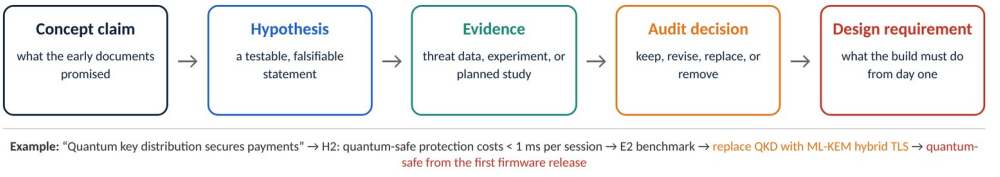


Figure 3.2 The claims-to-evidence method. Each claim moves left to right until it becomes a design requirement, or is removed.

Step 1. List every claim

Read your documents slowly and highlight every sentence that does one of four things: promises a result ("cuts wait times by 40%"), names a technology as the cause of a result ("using quantum algorithms"), states compliance ("meets FDA guidelines"), or gives a number ("$960,000 in savings"). Copy each one into a list, word for word, with

the document and page it came from. Do not fix anything yet. You are taking inventory.

When I did this for Glidecart and the related retail-robotics concepts, I found fourteen distinct claims across about twenty pages. Several appeared in more than one document, sometimes with different numbers, which is a warning sign in itself.

Step 2. Turn each claim into a hypothesis

A claim becomes testable when it has three parts: a **measurable outcome**, a **comparison**, and a **condition**. “Glidecart saves time” has none of them. “In a store with aisles, a cart that plans the shopper's route reduces walking distance by at least 20% compared with following the shopping list in order” has all three. Now you know what to measure, what to compare it with, and under what circumstances.

Good hypotheses can be wrong. That is the point. If no possible result could disprove your statement, it is a slogan, not a hypothesis.

Step 3. Find or create evidence

For each hypothesis, look for evidence and rate it on the ladder in Figure 3.3. The ladder has five rungs. The top rung is evidence you measured yourself with a method someone else could repeat. The next rung is a primary source, such as the official standard, the original study, a government filing, or the manufacturer's datasheet. Secondary sources, such as news articles, vendor blogs, and summaries of market reports, are useful leads but not proof. Below that are claims with no source, and at the bottom are claims that the evidence contradicts.

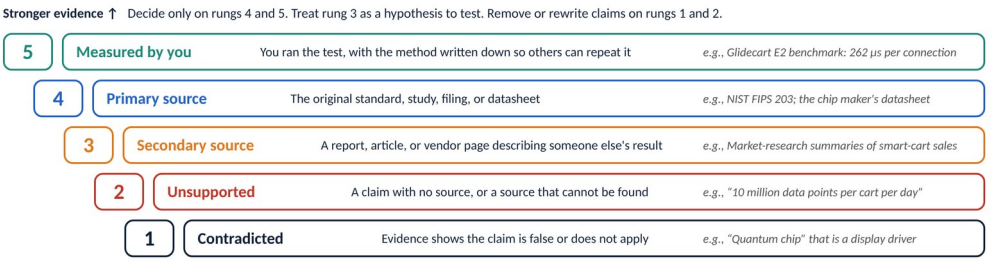


Figure 3.3 The evidence ladder. Make decisions only on rungs 4 and 5; treat rung 3 as a hypothesis; rewrite or remove anything on rungs 1 and 2.

If no good evidence exists, ask whether you can create it cheaply. Many technology claims can be tested on a laptop with free software. Others need people, which takes more planning (Chapter 11). Either way, writing down how you would test a claim is progress, because it tells you what you do not yet know.

Step 4. Decide: keep, revise, replace, or remove

Decision	Use it when	Example
Keep	Evidence on rung 4 or 5 supports the claim as written, or it is a clearly labeled hypothesis with a test plan	Deli kiosk wait-time target, kept as a hypothesis with a planned study
Revise	The idea is sound but the wording overstates the evidence	"Is compliant with" becomes "must be designed to meet"
Replace	The goal is right but the technology or cause is wrong	Quantum key distribution replaced by post-quantum encryption
Remove	The claim is false, cannot be supported, or should not be pursued	A "quantum chip" that is a display driver; biometric impulse detection

Step 5. Write the design requirement

Every decision should leave something behind that the builders must do. A replaced claim becomes a requirement to use the better method. A kept hypothesis becomes a requirement to measure it in the pilot. A removed claim often becomes a requirement *not* to do something, such as "the cart stores no audio and no images of faces." These requirements are the real output of the audit. They are cheap to write now and expensive to add later.

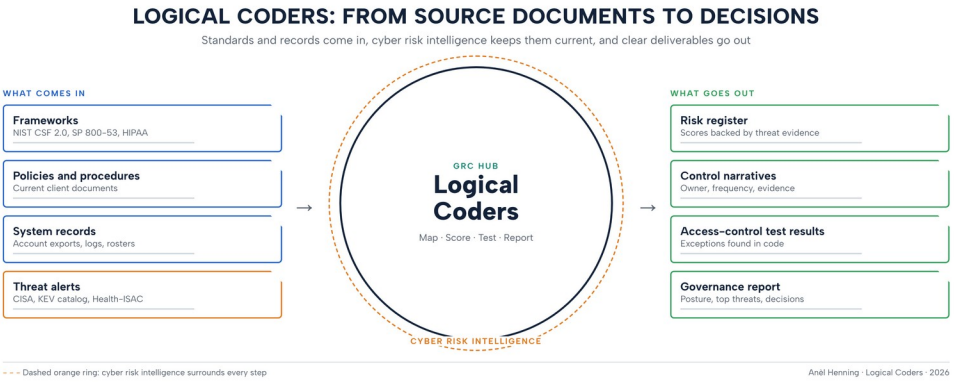


Figure 3.4 Source documents in, decisions out. The same flow that turns an organization's policies and records into a risk register and governance report also turns a product plan into decisions and design requirements. The orange ring marks where current threat intelligence informs each step (Chapter 4).

Worked Example: Fourteen Claims in the Glidecart Plans

Glidecart has six internal systems: a LiDAR sensor array, a vision module, a processing unit, a battery pack, two drive motors, and a wireless module (Figure 3.5). It has never been built. That makes it a good subject for this method, because every decision is still free to change.

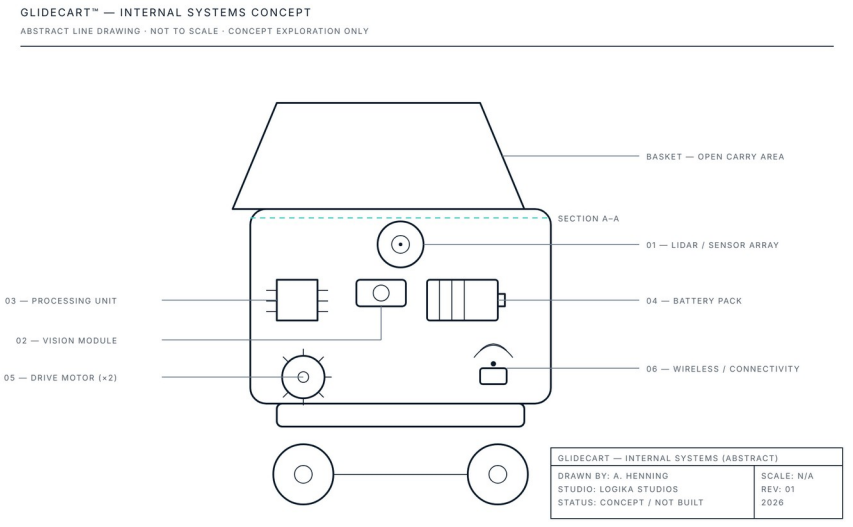


Figure 3.5 Glidecart internal-systems concept drawing. Abstract, not to scale, not built.

Table 3.1 at the end of this section lists all fourteen claims with their evidence and decisions. Here are the eight that taught me the most, grouped by the lesson they teach.

Lesson 1: Separate the benefit from the cause

The claim: "Quantum algorithms (QAOA) optimize the cart's route, reducing shopping time by 20%."

This sentence contains two claims. One is about a benefit: planning the route saves about 20% of the walking. The other is about a cause: quantum computing produces that saving. They need separate tests.

I tested both with a simulation of 30 randomly generated store layouts, each with an entrance, a checkout, and four items on different aisles. For each layout, an ordinary computer program checked every possible order of the four stops and picked the shortest. A quantum algorithm called QAOA, the one my plan named, was run on a simulator of a 16-qubit quantum computer on the same problems (Farhi et al., 2014).

The benefit held up. The best route was on average **18.4% shorter** than walking to the items in the order they appeared on the list, which is statistically indistinguishable from the 20% in my plan. The cause did not hold up. The ordinary program found the best route in all 30 layouts in about 0.06 thousandths of a second. The quantum algorithm found it in only **13 of 30** layouts and took about 38 seconds per layout. Most of its answers were not even valid routes.

What this means for your plan

The 20% saving is real, and you can get it for free from route-planning software that runs on a phone. Quantum computing added cost and nothing else at this scale. The correct decision is **replace**: keep the benefit, change the cause.

Ask of every claim with the word *because* or *using* in it: would the benefit still happen with a simpler, cheaper method?

Lesson 2: Check that the technology physically fits

The claim: "Quantum key distribution (QKD) encrypts the cart's transactions and shopper data."

QKD is a real technology. It uses individual particles of light to share encryption keys and needs dedicated optical links and special hardware at both ends. A shopping cart rolling around a grocery store has neither. The U.S. National Security Agency does not recommend QKD for protecting national security systems, citing its practical limits, and instead recommends new encryption algorithms designed to resist future quantum computers (National Security Agency [NSA], n.d.). NIST published those algorithms in 2024 (National Institute of Standards and Technology [NIST], 2024a, 2024b).

The goal behind my claim was right: protect payment and personal data against future quantum attacks. So I tested the replacement. On a laboratory computer, adding the new quantum-resistant method (ML-KEM) alongside today's standard method added about **262 microseconds** of processing per connection. For a modeled fleet of 200 carts each opening 300 connections a day, that is about **8 watt-hours a year**, less energy than a light bulb uses in an hour.

Decision: replace QKD with quantum-resistant encryption, and make it a requirement for the very first software release. Chapter 9 explains why waiting would cost far more.

Lesson 3: Look up every part number

The claim: the deli robots “leverage quantum chips such as the SSD7317Z.”

This took five minutes to check. The SSD7317 family, from the manufacturer's own product listing, is a *touch and display driver integrated* chip: it runs small touchscreens. It is an ordinary and useful part for a kiosk screen. It is not a quantum processor. The claim was on the bottom rung of the ladder: contradicted by a primary source.

Decision: remove. **Lesson:** whenever a document names a specific product, model, or part, look it up at the source. Names travel from document to document, and errors travel with them.

Lesson 4: Recompute every number

The claim: the deli kiosk saves “19,200 hours at $15 per hour, or $960,000 a year.”

Nineteen thousand two hundred times fifteen is **$288,000**, not $960,000. The input numbers may be reasonable; the result was wrong by more than three times. This kind of error is common in plans that go through many drafts, because the inputs get edited and the totals do not.

Decision: revise, and show the assumptions behind the 19,200 hours so a reader can check them. **Lesson:** put every calculation in a spreadsheet with its inputs visible. If a reviewer cannot recompute a number in under a minute, neither could you.

Lesson 5: Some claims create liabilities

The claims: “Each cart anonymizes over 10 million data points daily, including location, purchase history, and biometric inputs,” and a related concept for detecting shoppers' emotional “impulse” states with biometric sensors in order to time promotions.

The first claim had no source. More importantly, it described the cart collecting biometric data, which some laws strictly regulate. Illinois, for example, requires written consent before collecting biometric identifiers and lets individuals sue over violations (Biometric Information Privacy Act, 2008). The second concept had a deeper problem. Human-centered design asks *whom does this feature serve?* An impulse detector that times promotions serves the advertiser at the shopper's expense. The FTC has warned companies against exactly this kind of manipulative design (FTC, 2022).

Decision: remove both. In their place, the design now requires the cart to collect no biometric data, to process voice and video on the cart, and to keep only what it needs. The impulse idea became an opt-in budget assistant that shows shoppers their running total. It helps the person pushing the cart, and it needs no sensors at all.

Ask who the claim serves

For each claim, write down who benefits: the shopper, the store, the staff, the advertiser, or the investor. Claims that benefit only someone other than the user deserve extra scrutiny, and sometimes removal. This one question links human-centered design to risk management.

Lesson 6: You cannot be compliant with something that does not exist

The claim: “Materials are certified food-safe, and the cart is OSHA-compliant and ADA-accessible.”

A product that has not been built cannot have been certified or tested. These statements were intentions written as facts. They matter, because a buyer or regulator may rely on them.

Decision: revise to requirements: “The cart must be designed and tested to meet…” This small wording change turns an unsupported claim into a useful checklist for engineers, and it is honest.

Lesson 7: A foundational paper does not prove an applied claim

The claim: “NIST SP 800-208 supports QKD-based transitional cryptography.”

NIST SP 800-208 is a real and important document, but it covers a specific kind of digital signature, *stateful hash-based signatures*, which are well suited to signing software updates (Cooper et al., 2020). It is not about QKD. The same pattern appeared in Lesson 1, where my plan cited the original QAOA paper as if it had measured shopping time; that paper describes the algorithm and does not study stores at all.

Decision: revise, and use SP 800-208 for what it is good for: signing Glidecart's firmware updates so that only genuine updates can be installed. **Lesson:** when a citation supports a claim, open it and find the sentence that does so. If you cannot find that sentence, the citation does not support the claim.

Lesson 8: Good hypotheses are worth keeping

The claim: the deli ordering kiosk, with a screen that shows each order's status, “reduces wait times by 40%, increases satisfaction by 30%, and improves staff efficiency by 25% within eight weeks.”

Not every untested claim is a bad claim. This one is specific, measurable, and has a time frame. It just needs a test. Planning that test shows how affordable it is. If waits vary in the usual way, detecting a 40% reduction needs only about 20 orders with the kiosk and 20 without it; detecting a more modest 20% reduction needs about 98 of each. A busy deli serves that many orders in a few days.

Decision: keep, as a hypothesis with a written test plan (Chapter 11). This is the best outcome a claim can have before a product exists.

The Full Scorecard

Table 3.1 shows all fourteen claims. Of the fourteen, none was supported by a primary source or my own measurement when I first wrote it. Six were contradicted by evidence in whole or in part, five were unsupported or untested, one rested on secondary sources only, one had an arithmetic error, and one raised legal and ethical risk. After the audit, every row has a decision and every decision has a design requirement.

Table 3.1 Claims audit of the Glidecart and retail-robotics concept documents

#	Claim (paraphrased)	Evidence	Decision → requirement
1	Quantum routing cuts shopping time 20%	Benefit supported by my test; quantum cause contradicted	Replace → classical route planning
2	QKD secures payments	Contradicted: needs optical links; NSA does not recommend	Replace → post-quantum encryption from release 1
3	“Quantum chips such as the SSD7317Z”	Contradicted: display driver chip	Remove
4	Smart carts raise basket size 15%, cut scan errors 30%	Secondary sources only	Revise → measure in pilot
5	10 million data points per cart per day, incl. biometrics	Unsupported; legal risk	Remove → no biometrics, on-cart processing
6	Certified food-safe; OSHA- and ADA-compliant	Unsupported: not built	Revise → design and test to meet
7	SP 800-208 supports QKD	Contradicted: covers hash-based signatures	Revise → use it for firmware signing
8	Deli robots cut assembly time 50%, errors 40%, waste 60%	Unsupported for this product	Revise → pre-register as hypotheses
9	19,200 h × $15 = $960,000	Arithmetic error ($288,000)	Revise → show assumptions
10	Kiosk cuts waits 40% in 8 weeks	Untested; specific and measurable	Keep → planned field test
11	Biometric impulse detection for promotions	Legal and ethical risk	Remove → opt-in budget assistant
12	Grover's algorithm speeds product lookup	Contradicted: catalogs are indexed; Grover helps unstructured search	Replace → database index
13	Quantum recommendations personalize offers	Contradicted: a classical method matches the speed-up (Tang, 2019)	Replace → classical, opt-in recommender
14	Robots are energy-efficient by design	Untested	Keep → measure energy per trip

Seven Traps to Watch For

- **"Up to."** "Up to 30% faster" is true if one case out of a thousand was 30% faster. Ask for the typical result.
- **Round numbers with no method.** 20%, 30%, 50%, 10 million. Real measurements are rarely round. Ask how each was measured.
- **Market reports as proof.** A forecast that a market will grow says nothing about whether your product will work.
- **The famous-paper citation.** A landmark paper about a method rarely measured your application. Find the sentence that supports the claim (Lesson 7).
- **Copied part numbers and names.** Look them up at the manufacturer (Lesson 3).
- **Compliance in the present tense.** Unbuilt products are not compliant; they are designed to be (Lesson 6).
- **References that do not exist.** Drafts written quickly, whether by hand or with AI writing tools, can contain citations to articles, journals, or reports that cannot be found. In my own early documents, at least one cited journal article could not be located anywhere. Search for every reference by its title before you rely on it.

What the Audit Cost, and What It Saved

Auditing fourteen claims took me a few days of reading and checking, plus about an hour of computer time for the experiments, using free software. It did not require a lab, a quantum computer, or any purchases.

What it saved is harder to put in dollars, so I will not invent a number. What I can say is what it prevented. The plan no longer asks investors to fund quantum hardware that would not have helped. It no longer promises compliance it cannot yet prove. It no longer proposes collecting biometric data that would have created legal exposure. It no longer contains a revenue figure that was wrong by a factor of three. And it now contains about a dozen design requirements, each of which is far cheaper to build in today than to add after a fleet of carts is in stores.

Human-centered and honest

The most valuable result of the audit was not a number. It was a plan that serves shoppers and that I can defend, line by line, to an investor, a regulator, or a skeptical engineer. That is what *secure from the start* means.

Your Turn: Claims Audit Worksheet

Copy this table into a spreadsheet. Fill one row per claim.

Column	What to write
Claim	Exact words, with document and page
Type	Result · technology as cause · compliance · number
Who benefits	Shopper or user · staff · organization · advertiser · investor
Hypothesis	Measurable outcome + comparison + condition
Evidence and rung	Source and ladder rung 1–5
Test, if needed	Cheapest test that could prove it wrong; cost and time
Decision	Keep · revise · replace · remove
Design requirement	What the build must or must not do
Owner and date	Who will make the change, by when

Chapter 3 Checklist

☐ I listed every claim that promises a result, names a technology, states compliance, or gives a number.

☐ Each claim is rewritten as a hypothesis with an outcome, a comparison, and a condition.

☐ Each claim has an evidence rung from 1 to 5.

☐ I separated benefits from their supposed causes.

☐ I looked up every product name, part number, and citation at its source.

☐ I recomputed every number.

☐ I wrote down who benefits from each claim.

☐ Every claim has a decision, and every decision has a design requirement with an owner and a date.

☐ Untested claims I am keeping have a written test plan.

Try It

Exercise 1 (30 minutes). Pick a product page for a smart-home device, a point-of-sale system, or an AI tool your organization is considering. Highlight five claims. Rate each on the evidence ladder and write one hypothesis you could test.

Exercise 2 (60 minutes). Download the Glidecart experiment script from the book's companion repository and run the routing test on your own computer. Change the number of items from four to five. Does the classical method still win? How long does the quantum simulation take now?

Questions for Discussion

- Lesson 1 kept the benefit and replaced the cause. Can you think of a product you use whose advertised benefit comes from something simpler than the technology it names?
- Is it ever acceptable to publish a claim before it has been tested? How should it be worded?
- The impulse detector might have increased sales. Using the question *whom does this serve?*, argue for and against it.
- Which of the seven traps have you seen most often at work or school?

References

Biometric Information Privacy Act, 740 Ill. Comp. Stat. 14 (2008).

Cooper, D. A., Apon, D. C., Dang, Q. H., Davidson, M. S., Dworkin, M. J., & Miller, C. A. (2020). *Recommendation for stateful hash-based signature schemes* (NIST Special Publication 800-208). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-208

Farhi, E., Goldstone, J., & Gutmann, S. (2014). *A quantum approximate optimization algorithm* (arXiv:1411.4028). arXiv. https://doi.org/10.48550/arXiv.1411.4028

Federal Trade Commission. (2022). *Bringing dark patterns to light* [Staff report].

Federal Trade Commission. (2023a, May 31). *FTC says Ring employees illegally surveilled customers, failed to stop hackers from taking control of users' cameras* [Press release].

Federal Trade Commission. (2023b, May 31). *FTC and DOJ charge Amazon with violating children's privacy law by keeping kids' Alexa voice recordings forever and undermining parents' deletion requests* [Press release].

National Institute of Standards and Technology. (2024a). *Module-lattice-based key-encapsulation mechanism standard* (FIPS 203). https://doi.org/10.6028/NIST.FIPS.203

National Institute of Standards and Technology. (2024b). *Module-lattice-based digital signature standard* (FIPS 204). https://doi.org/10.6028/NIST.FIPS.204

National Security Agency. (n.d.). *Quantum key distribution (QKD) and quantum cryptography (QC)*. https://www.nsa.gov/Cybersecurity/Quantum-Key-Distribution-QKD-and-Quantum-Cryptography-QC/

Tang, E. (2019). A quantum-inspired classical algorithm for recommendation systems. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing* (pp. 217–228). ACM. https://doi.org/10.1145/3313276.3316310

Note on the cases: Glidecart is an unbuilt concept by the author. Sunshine Dental Group is a fictional organization. Experiment details and code appear in the author's research manuscript on which this chapter is based.